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Cylinder Block Machining Instruction for Water Pump Idler Capscrew Upgrade

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1.0 Overview

Water pump idler shaft joint has been redesigned to accommodate a higher preload. This document provides the information on how to incorporate a larger 3/4 in capscrew. Reference Figure 1 and Table 1 for parts familiarization and Figure 2 for existing drawing specifications.

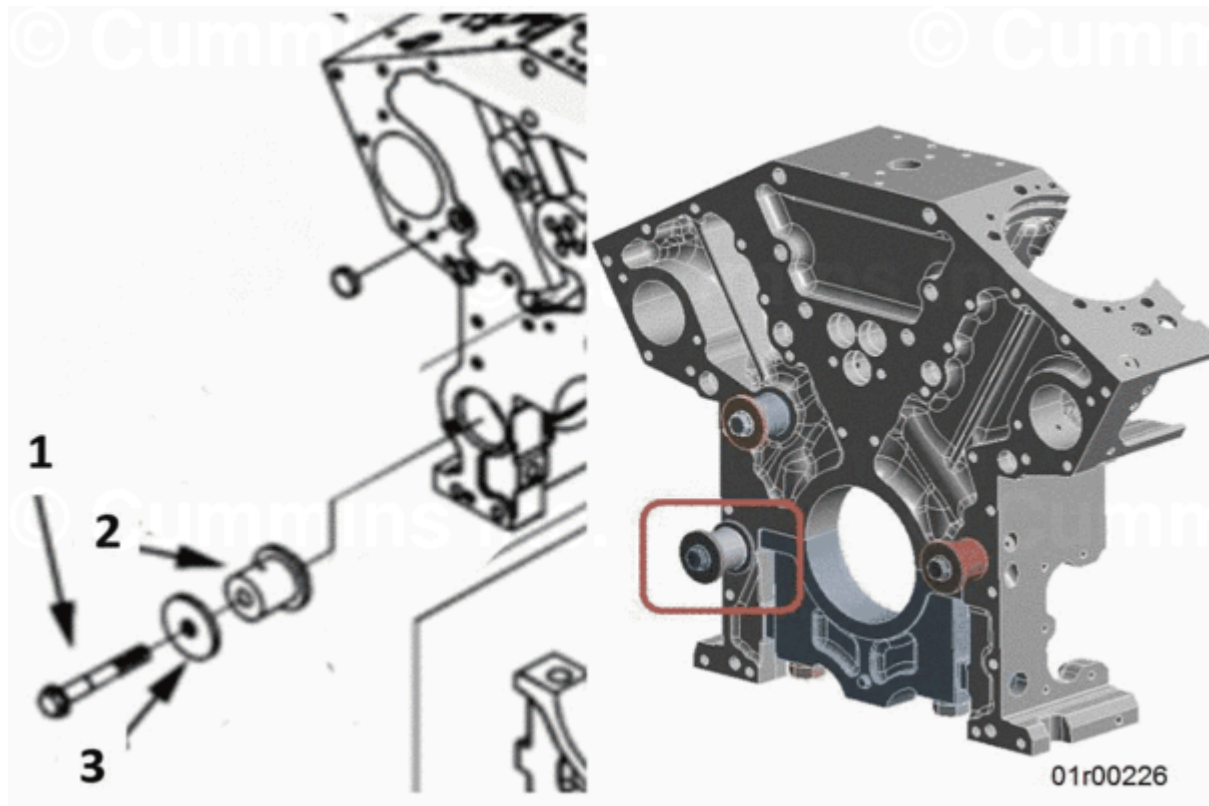


Figure 1, Part Familiarization

Table 1, Service Parts

Key	Part Description	5/8 in Hardware	3/4 in Hardware
1	Capscrew	3177443	3643905
2	Idler Shaft, Oversized length 1, Oversized length 2	3178450, 3178451, 3178452	3649522, 3649523, 3649524
3	Washer	3178457	3643906

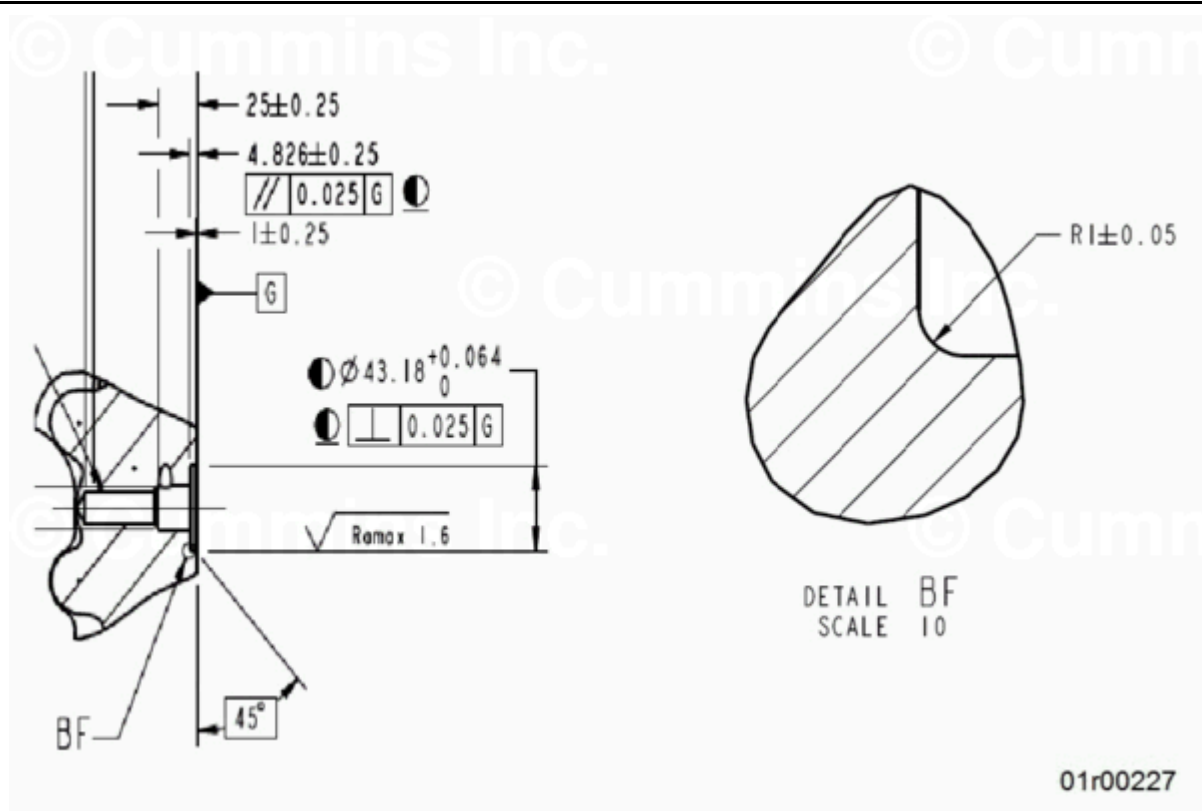


Figure 2, Drawing for standard 3/4 in capscrew.

1.1 Procedure selection

When up fitting new parts there are two available procedures:

Procedure A: Machining cylinder block in conjunction with an insert to restore the cylinder block back to factory dimensions.

Procedure B: Machining cylinder block to accommodate oversized idler shaft. (Oversized in length).

If ovality meets parameters in section 1.2, Procedure B can be performed. If counter bore does **not** meet ovality parameters, Procedure A **must** be performed.

1.2 Checking bore ovality

Procedure B requires existing ovality of bore to meet dimensions shown in Figure 4. This **must** be measured at three different depth points on counter bore shown in Figure 4. Tool can be rotated about each point A, B and C to check that ovality meets tolerance shown below.

If this measurement is in tolerance, either procedures A **or** B are acceptable. If bore is out of specification, **only** "Idler Bore Machining Procedure Procedure A" can be performed.

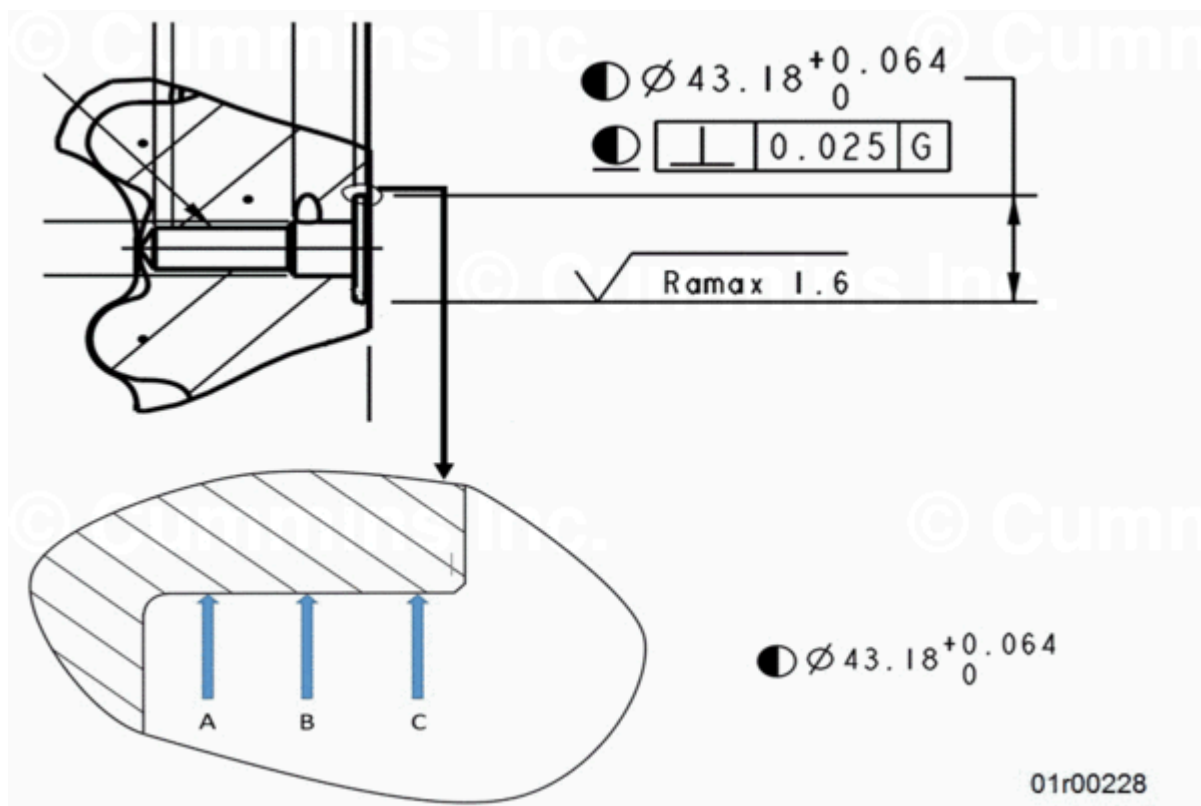


Figure 3, Ovality Parameters.

2.0 Idler Bore Machining Procedure Option A

2.1 Block alignment

Alignment of cylinder block is critical to ensure idler gear is correctly positioned relative to crankshaft and right bank camshaft idler.

Face of cylinder block should be cleaned without using abrasive material before cylinder block is aligned onto machine.

To check center of idler bore crankshaft center can be measured in reference to right bank camshaft idler bore center depicted in Figure 5. Using center of crankshaft and camshaft idler, target position must be within 0.1mm $\varnothing$ true position. This equates to a position tolerance of ± 0.05 mm on the measurements shown below.

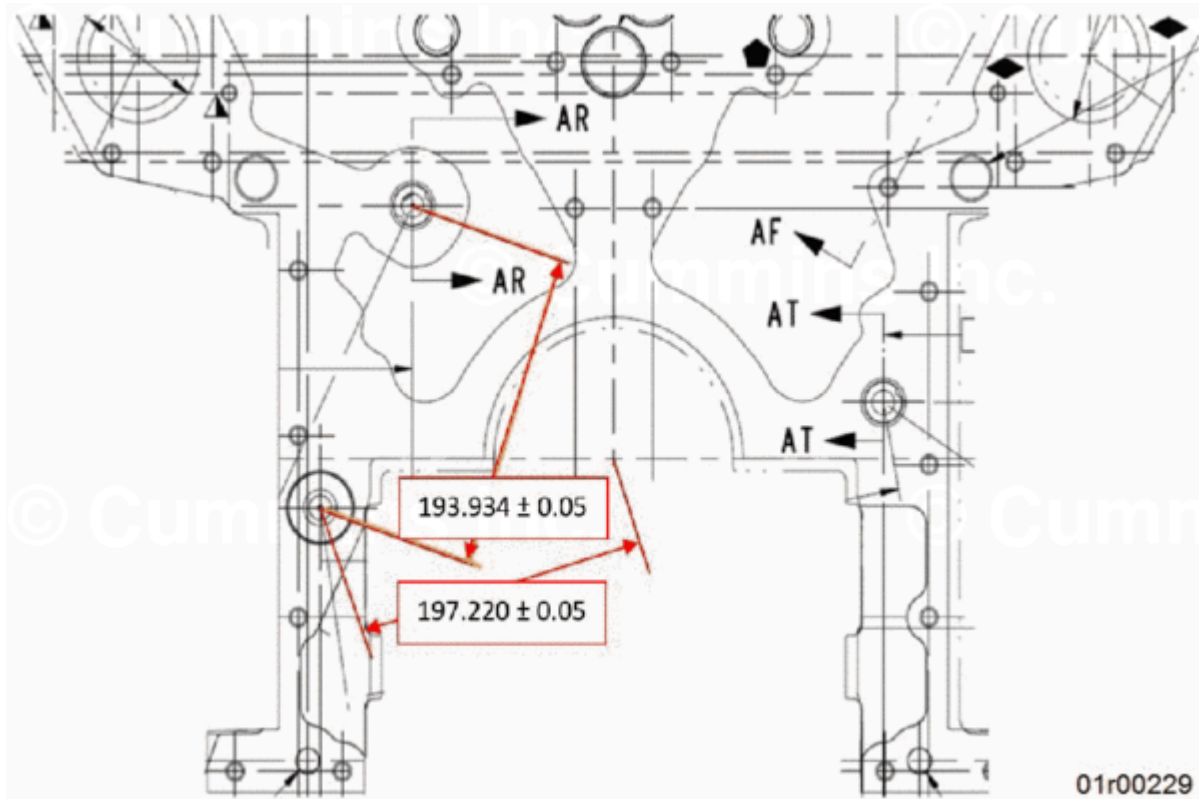


Figure 4, Alignment Referencing Crank Shaft Center and Right Bank Cam Idler Center.

In addition to alignment procedure, a crank alignment tool is available that can be used in conjunction with a stock idler.

Part Number 3375063 Crank centering ring (two required)

Part Number 3375062 Main bearing bore bar

After inserting crank center rings, torque mains to 68 N·m [50 ft-lb] holding the center rings in place. Insert the bearing bore bar into the centering rings. This bar can now be used as a measurement reference for the crankshaft center.

2.2 Initial block machining for insert

In order to remove ovality and surface defects, machine both circumference of bore and surface where idler contacts cylinder block. Figure 6 depicts exact width and depth of cut. This enables the cylinder block to accept an insert with intended interference fit.

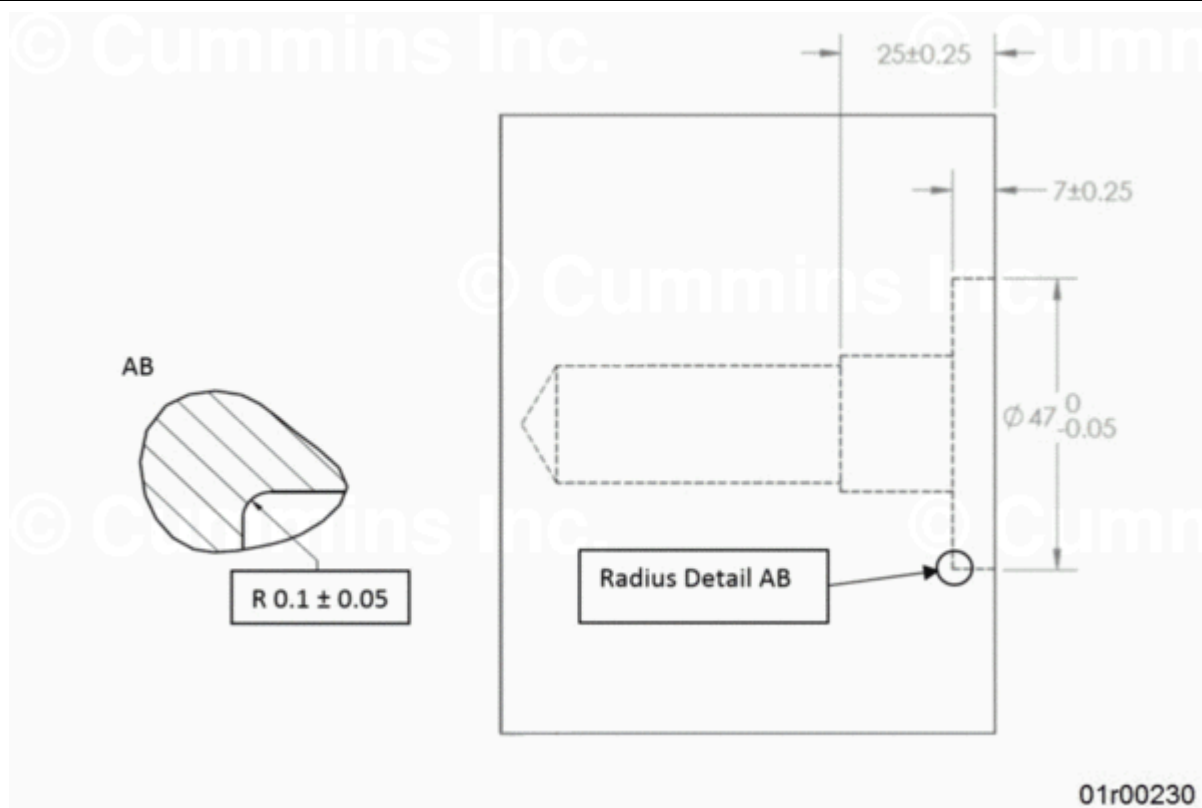


Figure 5, Machining 2.2.

2.3 Insert machining

Insert must be made from unalloyed cast iron, CES 40001. Use dimensions shown in Figure 7 to construct insert.

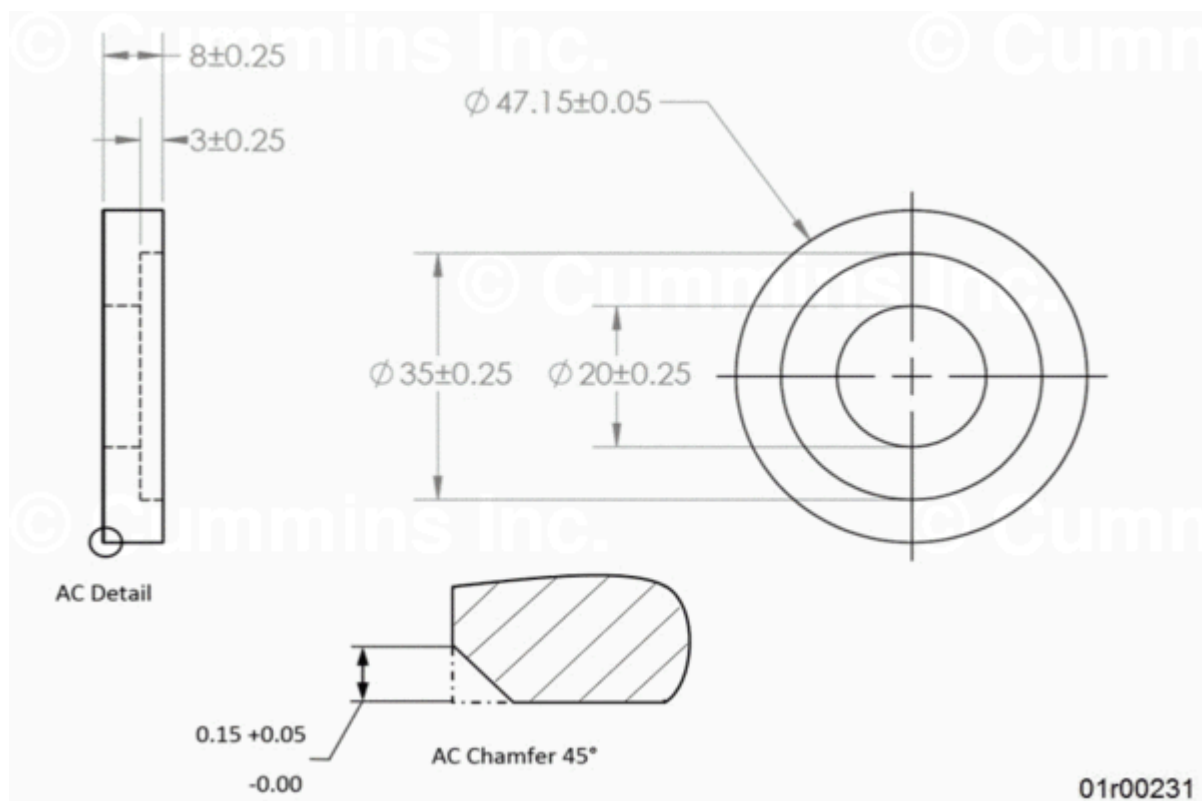


Figure 6, Insert Machining Dimensions.

2.4 Insert installation

Once machined tolerance of cylinder block and insert have been completed, both surfaces should be cleaned.

Sleeve **must** be cooled prior to installation.

Methods:

Freeze spray

Use an insulated container with two holes; one for entry of freeze spray and one as a vent. Place sleeve into container and apply spray until part has cooled sufficiently. Use appropriate personal protective equipment (PPE) to remove sleeve and any condensation.

Liquid nitrogen

Follow standard procedure for using liquid nitrogen interference fits.

Loctite™ 620/675 can be applied to idler bore prior to pressing insert. Temperature reduced insert should be pushed evenly into block. 2-3 hours **must** be allowed for Loctite to set before final machining. During Loctite curing time section 3.1 Capscrew hole and thread machining can be completed.

2.5 Final cylinder block face machining

Final Face tolerances shown in Figure 8 and Figure 9 in accordance with cylinder block drawing.

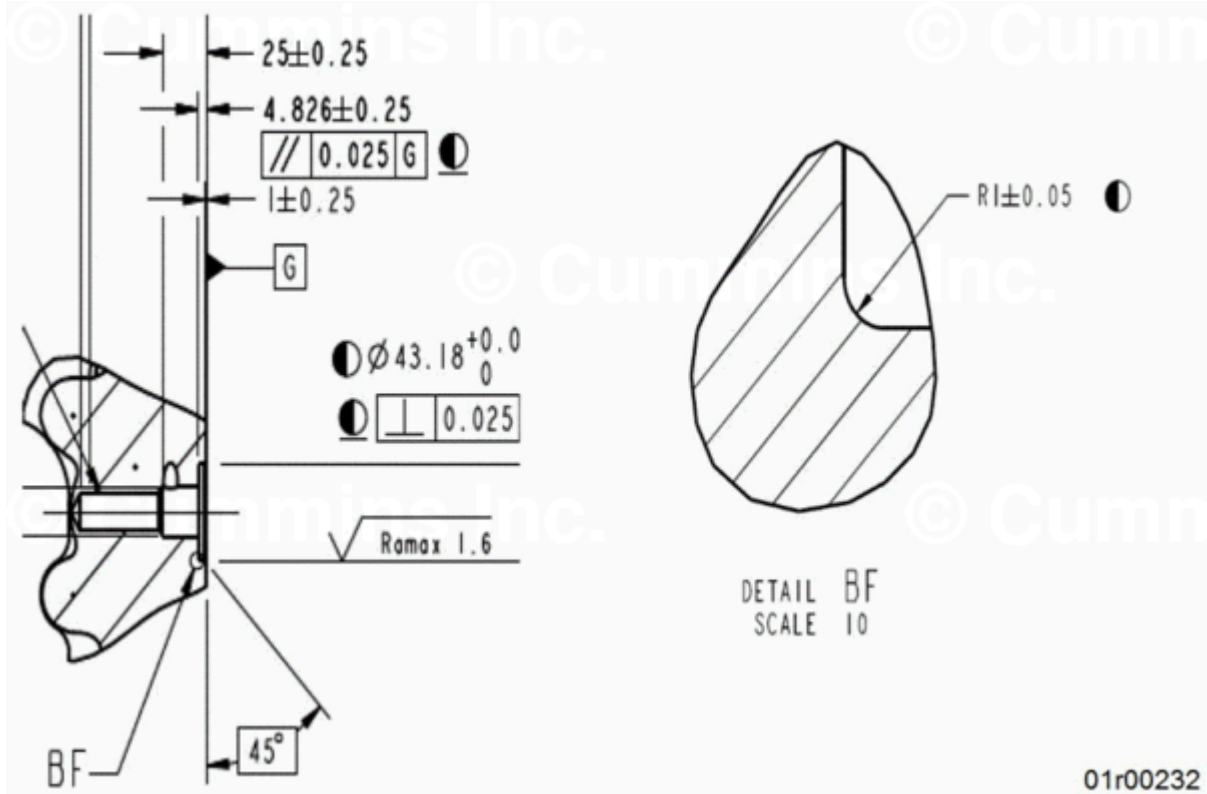


Figure 7, Final Face Machining.

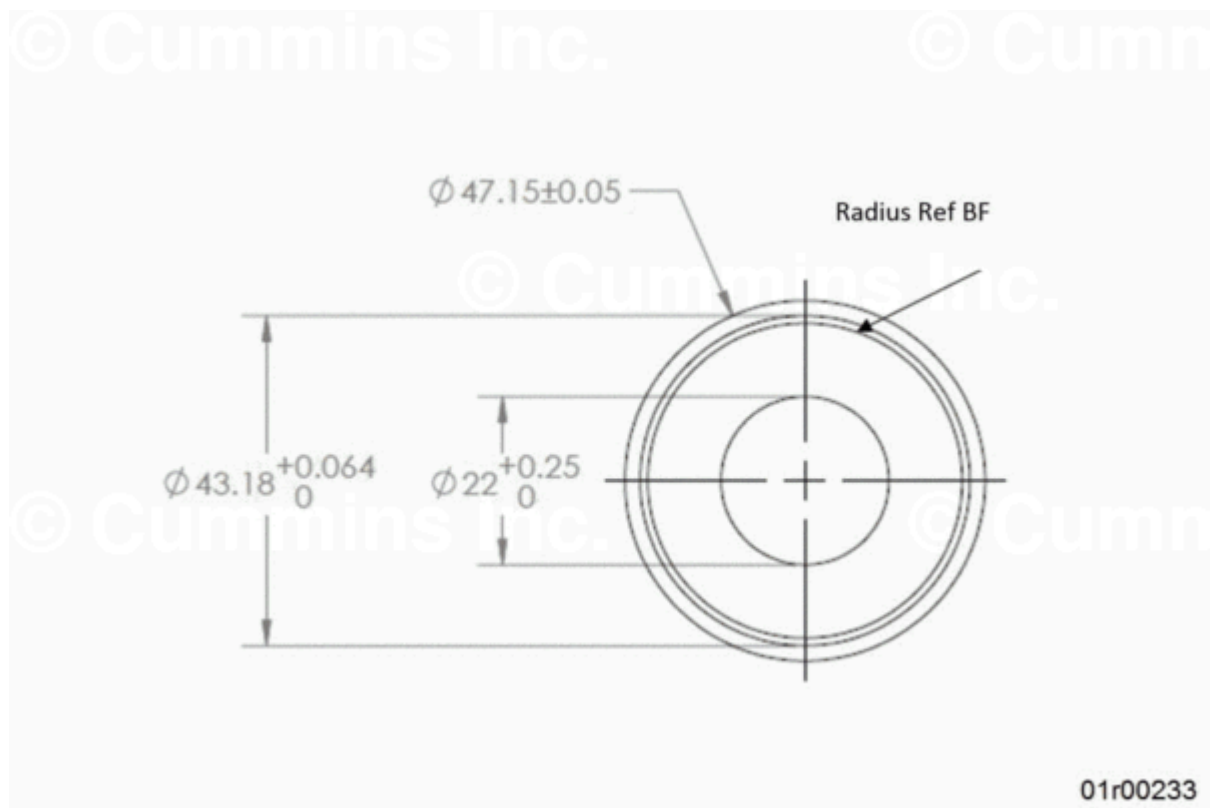


Figure 8, Final Face Machining With Respect to Insert.

3.0 Machining for $\frac{3}{4}$ UNF-2B Thread

3.1 Capscrew hole and thread machining

Following instructions shown in Figure 10, drill and tap new capscrew hole. Pilot hole is **not** to exceed 61mm to shoulder, and 69mm to tip. Take care not to exceed maximum depth specified. Always remove any debris that is generated in the process. Hole **must** be tapped to a minimum depth of 55mm.

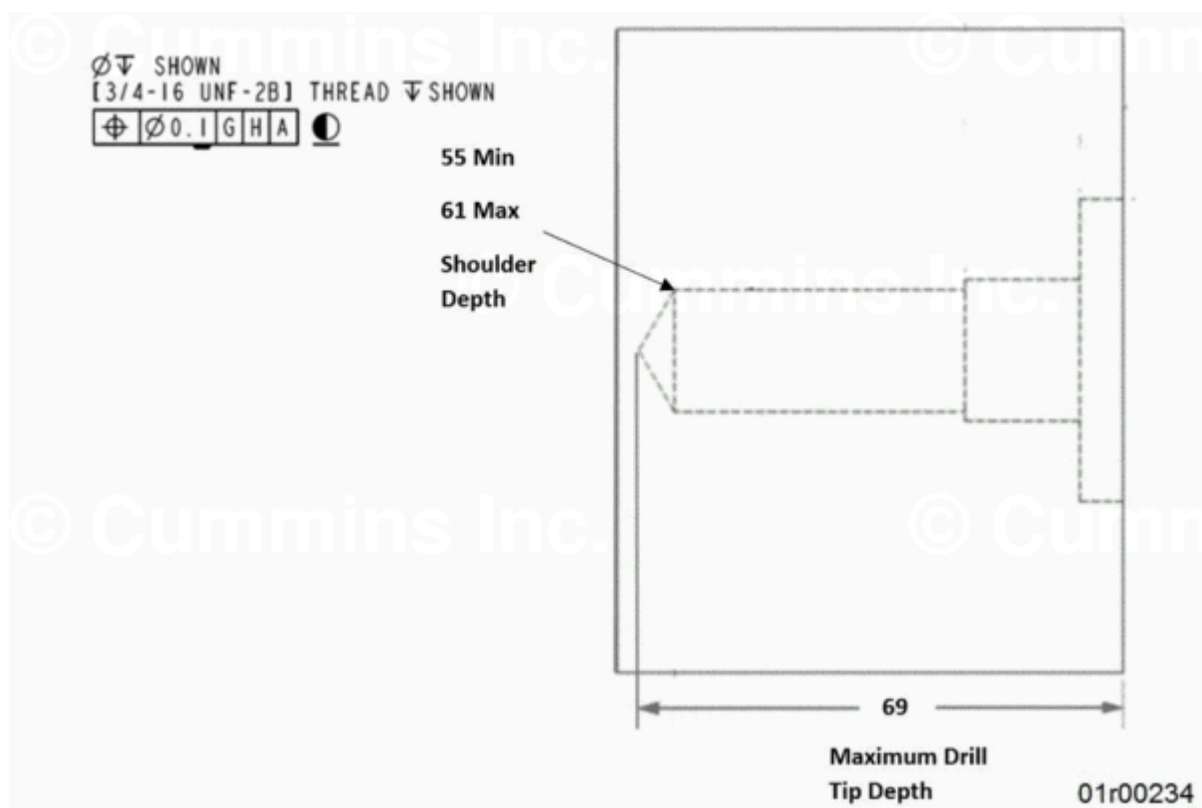


Figure 9, Capscrew Hole Width and Depth

4.0 Oversized shaft installation option B

4.1 Option B

Table 2 below depicts relevant bore depth for part number being selected. Reference Figure 11 for final face machining specification.

When centering tool in relation to the counter bore refer to section 2.1 Block alignment.

Refer to section 3.0 Machining for 3/4 UNF-2B Thread for thread machining.

Table 2, Part Number and length

Part Number	Dimension A
3649523	7.426
3649524	9.926

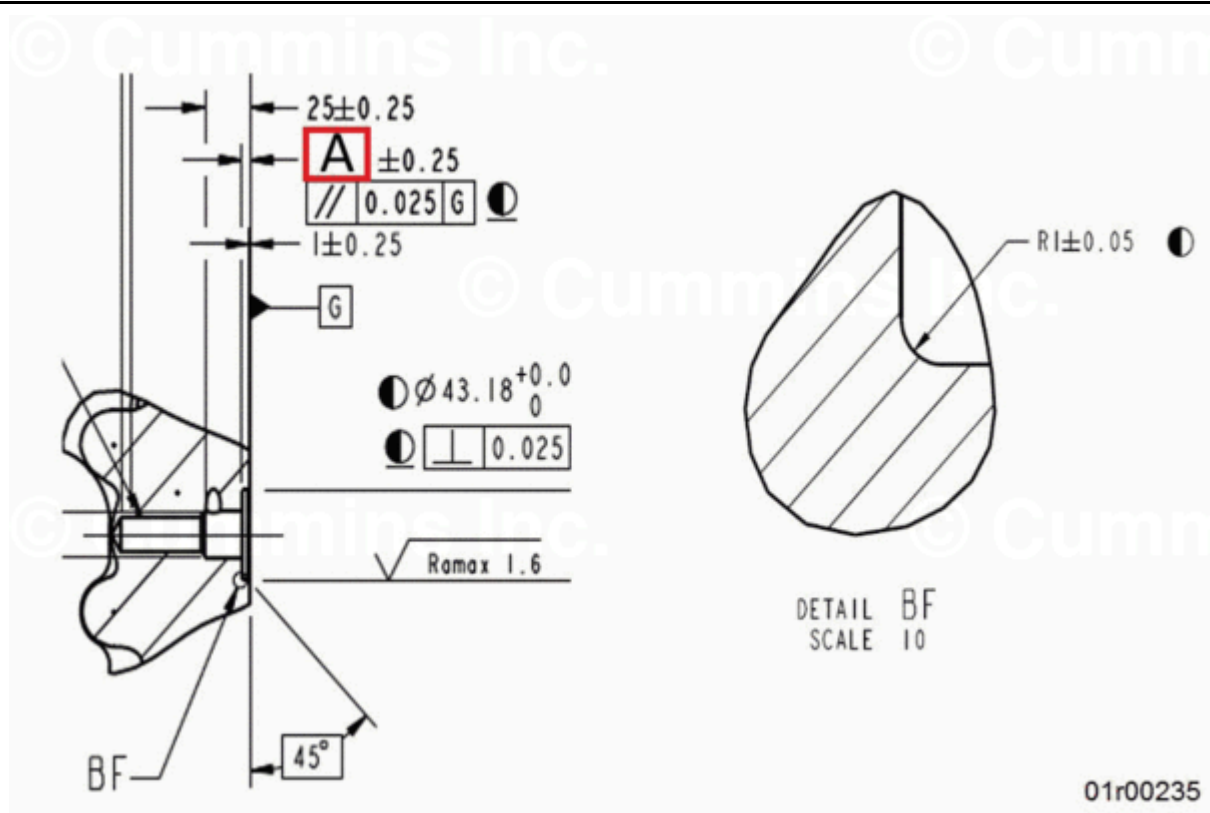


Figure 10, Final Face Machining and Counter Bore Depth for Oversized Shaft Options.

Document History

Date	Details
2018-7-11	Module Created

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